# OPERATION RESOLVE & TRACE: SOC ANALYST CTF CHALLENGE

**Course Module:** Network Investigations & DNS Reconnaissance

**Deliverable:** Individual Threat Intelligence & Reconnaissance Report (PDF)

**Submission Window:** 48 Hours

## SECTION 1: CTF OVERVIEW & RULES OF ENGAGEMENT

### Challenge Scenario

Your Security Operations Center (SOC) SIEM has raised alerts regarding anomalous network traffic and DNS queries generated by multiple hosts. As a Tier-1 SOC Analyst, you are assigned a specific target domain, public IP, and an internal private subnet segment to investigate. Your goal is to conduct DNS reconnaissance, verify infrastructure records, trace reverse DNS mappings, evaluate IP reputation, and perform authorized port validation.

### Required Investigation Tools & Commands

Utilize both native Linux CLI utilities and web-based OSINT threat intelligence tools:

1. **Linux CLI Utilities**:
    - **DNS Reconnaissance**: `dig` (A, AAAA, MX, NS, CNAME, PTR, +short, +trace), `nslookup`, `host`
    - **Host & System Context**: `hostname`, `ip a`, `ip r`
    - **Port & Exposure Validation**: `nmap` (`-sS`, `-sV`, `-Pn`, `-p`), `nc -zv`
2. **Web OSINT Threat Intelligence Portals**:
    - **Reputation & Threat Scoring**: VirusTotal - https://www.virustotal.com AbuseIPDB - https://www.abuseipdb.com/, https://www.ip2location.com/
    - **Geolocation & ASN Lookup**: IPLocation - https://www.iplocation.net/ Hurricane Electric BGP/DNS - https://bgp.he.net/, https://dnschecker.org/

### Required Report Structure

Submit a report containing:

1. **Executive Summary**: Target assignment, high-level findings, and risk assessment score.
2. **DNS Record Matrix**: Exact records found (A, AAAA, MX, NS, CNAME, PTR) using `dig`/`nslookup`.
3. **OSINT & Geolocation Analysis**: ISP, Country, ASN, Abuse Score, and VirusTotal detection flags.

4. **CLI Execution Screenshots / Terminal Output**: Code snippets showing commands executed.
5. **Mitigation / SOC Recommendation**: Actionable steps based on findings.

# Assignment Details

- **Target Domain**: `isc.org` (Internet Systems Consortium)
- **Target Public IP**: `204.152.184.88`
- **Target Private Subnet**: `172.31.1.50/24`
- **Tasks**:
    1. Query `isc.org` for A, AAAA, MX, and NS records.
    2. Perform PTR lookup for `204.152.184.88`.
    3. Evaluate `204.152.184.88` on VirusTotal and AbuseIPDB.
    4. Compare output of `dig isc.org MX` vs `nslookup -type=MX isc.org`.